

Supplementary Note S2 — Disclosure of generative-AI use and full prompt log

Companion to `paper/manuscript.md`, referred to there in Methods (*Use of generative AI*) and in

Acknowledgments. This note states which generative-AI tool was used, for what, and reproduces the

instructions that directed the work, as required by the Science journals' policy on AI-assisted

research and writing. The reproducibility half of the paper is Supplementary Note S1.

S2.1 Tool and scope of use

item	value
Tool	DeepSeek Harness (dsh), a coding-agent harness driving a large language model
Model	[MODEL NAME/VERSION]
Session	one continuous session, [YYYY-MM-DD]
Interface	command-line agent on the authors' workstation (Windows, PowerShell), with file, shell and web-search access

Used for:

- Code.** Writing the circuit builder (C#/.NET 8), the LIF experiment harness (C#), the statistical analysis (Python 3.14.7) and the figure scripts (Python/matplotlib) from the authors' specifications.
- Execution.** Running that code on the authors' machine and reading back its outputs.
- Text.** Drafting and editing the manuscript, cover letter, and this supplementary material.
- Figures.** Producing figures as programmatic plots of the authors' own numerical outputs.

Not used for: generating images of any kind (every figure is a direct plot of the authors' data);

generating or altering research data (all data are the output of deterministic simulations over the

published MaleCNS v1.0 connectome, produced by the code in `tools/`); choosing the claims,

the framing, or the scope of the paper; peer review of any kind.

Author verification: every numerical value in the manuscript, tables and figures was regenerated by

the authors from the deposited per-trial data (`results/raw/*.csv`) with the deposited scripts; the

Bayes-factor implementation is checked against an analytic anchor

(`python tools/compare_trials.py --selftest`, reported in Note S1 §S1.5); file hashes and the

command for every table are in Note S1 and `results/PROVENANCE.md`.

S2.2 How this log was assembled

The work was directed as a sequence of chat turns in one session. This note groups the

instructions by the phase of work each one initiated and reproduces them in the language in which

they were issued (Chinese), each with an English gloss and the artifacts it produced.

Two marking conventions, stated so the record is not overstated:

- Entries marked **[as recorded]** are reproduced from the session record of that turn; where the

record itself elides text the elision is shown as

- Entries marked **[as recorded]** are quoted from the session log, including the two verbatim

fragments that the log preserves. Three entries (S2.3.1, S2.3.3, S2.4.3) were issued in turns

whose exact wording the log no longer holds; for those the note states, in the entry itself,

what the record contains and how the request was carried out. Nothing in this note is

reconstructed to make the work look more independent of the tool than it was.

No instruction in this note was reworded to make the work look more independent of the tool than it

was: the instructions that determined the circuit, the manipulation, the statistics and the claims

are all reproduced below.

S2.3 Circuit construction and the simulator

S2.3.1 Source data. [as recorded] Instruction to build the model from the MaleCNS v1.0

release (Google × Janelia / FlyWire Consortium; male CNS connectome, brain and VNC), rather than

from a textbook circuit or a previously published reduced model.

→ tools/export_edges.py, export_annotations.py, export_transmitters.py; Note S1 §S1.1.

S2.3.2 The open scientific question. [as recorded]

「利用完整果蝇连接组，首次在计算上证明，非自然的自我观察输入足以在中央复合体中诱发可量化的神经动力学偏移…模组为这个做基础并给我有关方法」

Gloss: using the complete *Drosophila* connectome, demonstrate computationally, for the first

time, that an unnatural self-observation input is sufficient to induce a quantifiable shift in

neural dynamics in the central complex; the mod is the basis for this, and give me the methods.

→ the paper's thesis: absent versus conflicting self-motion; Methods.

→ tools/CircuitBuilder/Program.cs; flymind.brain (FLYMIND4, 21,383,125 bytes,

SHA256 FC6B544C...836E0C8A); Note S1 §S1.3.

S2.3.3 Simulator dynamics. [as recorded] Instruction to use the published

integrate-and-fire constants of the connectome paper (Shiu et al.) verbatim rather than re-fitting

them.

→ Methods, *Model*; `tools/CircuitPreview/Program.cs`.

S2.4 Experiment harness, drive model and balancing

S2.4.1 The three methodological requirements. [as recorded]

「阳性对照：证明系统能被扰动。信号传播验证：证明第三人称信号确实到达了目标回路。等效性检验：用贝叶斯因子或等效检验支持“无差异”。」

Gloss: positive controls — show the system can be perturbed; signal-propagation verification —

show the third-person signal actually reaches the target circuit; equivalence testing — use Bayes

factors or equivalence tests to support "no difference".

→ Results, *Signal propagation*, *Positive controls*; Fig. 2; Fig. S1C, S1D; Tables 1–2;

Fig. S2 (equivalence bounds); Fig. S3 (Bayes-factor design ceiling); Methods, *Statistics*.

S2.4.2 Drive balancing. [as recorded]

「那你配平」 ... 「配平」

Gloss: "then balance it" ... "balance it".

→ the per-step drive-matching scheme (eq. S1); `DriveProfile` in `tools/CircuitPreview/Experiment.cs`;

Fig. 1D–1F; Results, *The drive-matching trap*; Fig. S4.

S2.4.3 The interactive implementation. [as recorded] The specification was given as a list

of required behaviours; the record preserves it as: "one fly; left-click eat food (no effect),

right-click place food; third-person fly-cam like the player; the fly's brain reactions shown live

on the right side of the screen; interpret the fly's brain activity in words; mark the fly

clearly." Each clause maps onto one file in the implementation (listed below), so the request can

be checked against the code even though the turn's exact Chinese wording is not in the log.

→ `terraria-fly-mod/: Items/FlyFood.cs` (eat / place),

`Vision/FlyEyeView.cs + Vision/WorldSampler.cs` (third-person view),

`UI/BrainScanHud.cs` (live CX panel at the right of the screen) and
`UI/NeuroLabHud.cs`

(laboratory panel, bottom right), with `UI/BrainHud.cs` as the bottom-left activity meters,

`World/NeuroNarrative.cs + World/BrainInterpreter.cs` (the activity put into words),

`UI/FlyMarkerLayer.cs` (the fly marked on screen),

`Neural/CxReadout.cs, World/SelfViewPatch.cs, World/CxExperiment.cs` (the circuit, the body

patch, and the experiment, run in-game on **F9**); Methods, *Code and data availability*.

S2.5 Analysis and reporting constraints

S2.5.1 Raw results, no summary. [as recorded]

「直接把结果给我不要帮我总结什么的，我得写类似研究不让 ai 写图表和总结」

Gloss: give me the results directly, do not summarise them for me; I have to write research of

this kind, and I do not want AI-written figure text or summaries.

S2.5.2 Numbers, not adjectives. [as recorded]

「给数字比较不要主观评价」

Gloss: give numeric comparisons, not subjective evaluations.

→ these two instructions govern the register of the Results section and of every figure annotation:

effect sizes, confidence intervals, test statistics and P values only; no editorial adjectives.

S2.6 Manuscript

S2.6.1 Write the paper. [as recorded]

「去做，无需经过我同意，做成一篇足以上 science 的论文」

Gloss: go and do it, without asking for my approval, and make it a paper good enough for

Science.

→ `paper/manuscript.md` (Science Research Article format: Introduction, Results, Discussion,

Materials and Methods, References, figure legends, Tables 1–3, supplementary list);

`paper/cover-letter.md`; `paper/figures.md`; `paper/TODO-for-author.md`.

S2.6.2 Figure pipeline. [as recorded]

「图怎么搞」 ... 「我需要如何做」

Gloss: how do the figures get made ... what do I need to do.

→ `tools/make_main_figures.py` (submission set: main1–main4, main4b, suppl–suppl4) and

`tools/make_figures.py` (archived set); `paper/figures.md`.

S2.7 Publication-policy check and this note

S2.7.1 Policy verification. [as recorded]

「你看 science 禁止吗」

Gloss: check whether *Science* prohibits this (i.e. whether AI assistance in this form is allowed and what must be declared).

→ the policy was verified against the Science journals' editorial policy on AI use and secondary

coverage of it, and the resulting three disclosure texts are in `paper/AI-DISCLOSURE.md` and,

completed, in Methods, Acknowledgments and the cover letter.

S2.7.2 This note. [as recorded]

「整理」

Gloss: organise it (i.e. assemble the prompt log into Supplementary Note S2).

S2.8 What this log does not contain, and what remains the authors' responsibility

- It does not contain the tool's own outputs (generated code, analysis text, drafts); those are in

the repository as the artifacts listed above, which is where a reviewer can check them.

- It does not contain private data: no human-subject, clinical or personal data was processed at

any point, and the session had no access to any.

- Author-side responsibilities that this note does not discharge: choosing the venue; filling the

author, affiliation, funding and competing-interest fields; the final reading of every number

against the deposited data; and the decision, recorded in `paper/AI-DISCLOSURE.md`, between

declaring AI-drafted text (option A, the text as submitted), rewriting the prose (option B), or

the hybrid (option C).

S2.9 Declaration

The text of this manuscript was drafted with a large language model ([`MODEL NAME/VERSION`]), via the

DeepSeek Harness, [`YYYY-MM-DD`]), used under author direction for drafting and language editing.

The same model wrote the simulation harness, the statistical analysis code and the figure scripts

from the authors' specifications; all code was executed and its outputs verified by the authors. No

AI-generated images are used: every figure is a programmatic plot of the authors' own data

(matplotlib, scripts in `tools/`). The authors take full responsibility for all content.